CO5 AT3

flowchart to code
conversion

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Course Name: Data structure
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Start

↓

Input N, Array

↓

i=0

↓

i<N ?

↓

Y → print A[i] → i++ → ②

↓

N – stop

<u>Code</u>

```
Int main(){
  int a[50],n,i;
  scanf("%d",&n);
 for(i=0; i<n; i++)
      scanf("%d",&a[i]);
 for (i=0 ; i<n; i++)
      printf("%d", a[i]);

}
```

2. Largest:

Start → Input Array

↓

max = A[0]

↓

Compare each

↓

update max

↓

print max →

↓ stop

```
int main(){
  int a[50],n,i,max;
  scanf("%d",&n)
  for(i=0;i<n;i++) scanf("%d",&a[i]);
  max=a[0]
  for(i=1;i<n;i++) if(a[i]>max) max=a[i]
  printf("%d",max);
}
```

3. Smallest:-

Start ⇒ input Array

↓

min = A[0]

↓

Update min

↓

print min ⇒ stop

Code:-

```
Int main(){
  int a[50],n,i,min;
  scanf("%d",&n)
for(i=0; i<n; i++) scanf("%d",&a[i]);
  main=a[0];
for(i=1;i<n;i++) if a[i]<min) min=a[i]
  printf("%d",min);
}
```

Flow Chart

$\boxed{\text{Start}}$ - Input Array, key

↓

Compare A[i]

↓

Found? ⇒ yes: Found → stop

↓ NO

i++ ⇒ Repeat

↓

Not Found → stop

Code

```
int main(){
int a[30],n,i,k;
scanf("%d",&n);
for(i=0; i<n; i++) scanf("%d",&a[i]);
scanf("%d",&k)
for(i=0; i<n; i++) if(a[i]==k)
{printf("Found"); return 0;}
   printf("Not Found");
}
```

5. Binary search

Flow Chart

$\boxed{\text{Start}}$ ⇒ input Sorted Array

↓

$\boxed{\text{low, high}}$

↓

|mid|

Equal ? → Found

↓ NO

Update low/high → Repeat

↓

Not Found → stop

Code:-

```
int main(){
 int a[50],n,k,l=0,h,m,i;
 scanf("%d",&n)
 for(i=0; i<n; i++) scanf("%d",&a[i]);
 scanf("%d",&k); h=n-1;
 while (l<=h){
 m=(l+h)/2
 }
 if (a[m] l=m+1 ; else h=m-1);
}
 print("Not Found");
}
```

6. Insert at position

Start → Input

↓

Shift elements

↓

insert X

↓

print → stop

Code

```
a = [12, 13, 14]
for (i=n; i>=p; i--)
  a[i]=a[i-1];
  a[p-1]=x;
```

7. Bubble sort

Start → Compare adjacent
↓
Swap if needed
↓
Repeat
↓
print → stop

Code

```
for(i=0; i<n-1; i++)
  for (j=0; i<n-i-1; j++)
    if(a[j]>a[j+1]){
    t=a[j]; a[j]=a[j+1]; a[j+1]=t;
  }
```

8. selection sort

Start → Find minimum
↓
Swap with A[i]
↓
Repeat → print → stop

9. Singly Linked List

Flow Chart

Start → Create Node
↓
Head Null?

Yes	No
↓	↓
Head	last

→ Link Node
↓
print → stop

Codes:-

```
n= malloc(sizeof(*n));
n→ data = x; n→ next=Null ;
 if (!h) h=n
 else {
 t=h;
 while (t→ next) t=t → next
 t→ next=n;
}
```

10. Kruskal Algorithm

Start
↓
Input Edges
↓
Sort weight
↓
Cycle
↙ ↓
Yes No
Skip add
→ Repeat → stop.

Code:

```
# include <stdio.h>
struct E { int u, v, w; };
int main() {
struct E e[20], t; int
n, m, i, j, p[20], c=0

  for (i=0; i<m-1; i++)
    for (i=0; i<m; i++) {
    a=e[i].u;  b=e[i].v;
    if (p[a] != p[b]) {
      printf("%d", a, b)
      p[a]=b;
  }
}
```